

C. Sergio Vavra

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EDUCATION

James Madison University, B.S. Computer Science - Harrisonburg, VA 2026

- 3.813 GPA, 5x Presidents list, 6x Deans list

Deep Learning Specialization & Machine Learning Coursera Certificate 2024

- Completed comprehensive coursework covering foundational concepts of machine learning, neural networks, and deep learning architectures, including convolutional and recurrent networks.

PROFESSIONAL EXPERIENCE

AI/ML & Backend Software Engineer Intern - Ellucian, Reston VA 2025 - Current

- Led the AI/ML development for a cross-functional team building an Agentic product, leveraging AI frameworks like LangChain, LangGraph, and AWS Bedrock to design models and systems.
- Designed and implemented scalable backend systems supporting agentic workflows, vector search integration, and RAG pipelines, optimizing performance and deployment across AWS services.

Site Manager - University Recreation Center, JMU, VA 2024 - Current

- Led and coordinated 70+ diverse teams, delegating responsibilities and maintaining operational efficiency in a fast-paced, dynamic environment.
- Managed a staff of 12, demonstrating strong leadership, conflict resolution and communication.

Computer Science Student Researcher - JMU, VA. 2023 - 2024

- Developed and implemented a Rasa chatbot leveraging AI and machine learning methodologies as part of a research project under Professor Isaac Wang at JMU
- Designed and customized chatbot actions utilizing python, demonstrating proficiency in leveraging programming languages for tailored solutions in AI-driven applications using machine learning

Database and Software Intern - Hiring Our Heroes, VA Summer 2022

- Conducted quality assurance (QA) by systematically reviewing websites to ensure functionality and optimal performance, reporting any suboptimal findings.
- Oversaw the management of a high-volume photo database, executing efficient CRUD operations.

PROJECT EXPERIENCE

Agentic AI Framework Odyssey - Professional Project, Ellucian 2025- Current

- Developed a personalized Agentic AI system for personalized career advice, leveraging RAG pipelines, vector embeddings, and multi-agent orchestration for contextual, adaptive recommendations.
- Streamlined retrieval and prompt orchestration using LangGraph for agent logic, LangChain for RAG pipelines, and LangSmith for evaluation and tracing across AWS Bedrock-hosted LLMs.

Professional Website with AI Integration - Personal Project, JMU 2024 - Current

- Developed a serverless chatbot on AWS using ECS, Lambda, API Gateway, and CloudWatch, enabling dynamic task management and integration with a custom front-end interface
- Built a personal website to host an AI chatbot using HTML, CSS, and JavaScript to showcase experience.

Multi-Functioning Calculator - CS 345 Final Project, JMU Fall 2023

- Leveraged Java programming language to architect and develop core functionalities of a multi-function calculator, ensuring robustness and efficiency in computational operations
- Led Scrum implementation, daily stand-ups, sprint planning, and retrospectives to meet milestones.

ActivityMixr - CS 343 Semester Project, JMU 2023 - 2025

- Developed and integrated multiple RESTful APIs to power an interactive web app that generates personalized playlists based on user-selected activities.

TECHNICAL SKILLS

Software - LangChain, LangGraph, AWS, Docker, TensorFlow, Poetry, Linux, Unix, GitHub, AWS SDK

Programming Languages - Python, TypeScript, Java, C, Ruby, Rust, Haskell, HTML, CSS, JavaScript, SQL, R.